1. Introduction

In modern smart homes, automation plays a vital role in improving safety and comfort. Environmental factors such as strong wind can cause damage when doors and windows are left open. This project proposes a Smart Wind Detection System that automatically controls smart doors and smart windows based on wind conditions. Using IoT technology and simulation, the system detects wind and performs automatic actions to protect the home.

The system uses an ESP32 controller connected to a wind detection sensor and output indicators. When strong wind is detected, the system automatically activates safety responses and provides real-time monitoring through wireless communication.

2. Objectives

- To design an automated system for door and window control
- To detect wind conditions using a smart wind detector
- To prevent damage caused by strong wind
- To implement the system using IoT concepts
- To simulate the system using ESP32 and IoT environment

3. Problem Statement

In traditional homes, doors and windows are manually operated. During sudden strong wind, users may forget to close them, leading to property damage, dust entry, and safety risks. There is a need for an automatic system that can detect wind conditions and control doors and windows without human intervention.

4. Proposed System

The proposed system uses a wind detection sensor connected to an ESP32 microcontroller. When wind intensity exceeds a predefined threshold, the system automatically activates safety indicators representing door and window closure. The system connects to WiFi and sends real-time data for monitoring and control.

5. System Architecture

The system architecture consists of:

- Wind detection sensor
- ESP32 microcontroller
- Smart door and window indicators (LEDs)

• Buzzer for alert notification

• WiFi communication module

All components communicate through the ESP32 controller, ensuring real-time response and wireless monitoring.

6. Hardware / Software Requirements

Hardware Requirements

• ESP32 Development Board

• Wind Sensor Module

• LEDs (Door and Window Indicators)

• Buzzer

• Resistors

• Connecting Wires

Software Requirements

• IoT Simulation Environment

• ESP32 Programming Environment (Arduino IDE)

• Serial Monitor

• Firebase Realtime Database (Cloud Data Monitoring Platform)

• WiFi Network

7. Methodology

1. Configure ESP32 microcontroller
2. Connect wind sensor to ESP32 input pins
3. Connect LEDs and buzzer as output devices
4. Program threshold-based automation logic
5. Establish WiFi connection
6. Monitor system behavior through serial output

8. Implementation

The system is implemented using ESP32-based IoT simulation.

• The wind sensor continuously monitors environmental conditions

• LEDs represent smart door and window status

• Buzzer provides alert during strong wind

• ESP32 processes sensor input and executes control actions

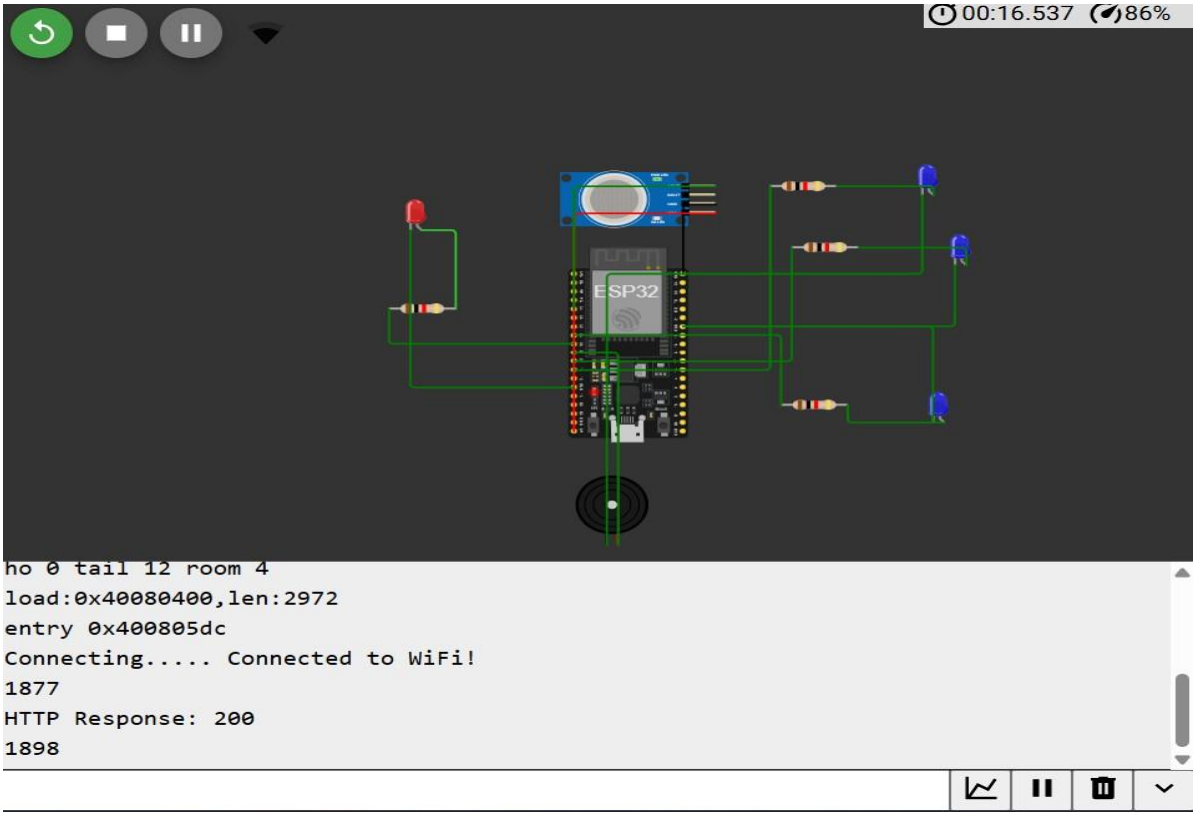
• WiFi connectivity enables real-time monitoring

This simulation represents a practical smart home automation setup.

9. Working Principle

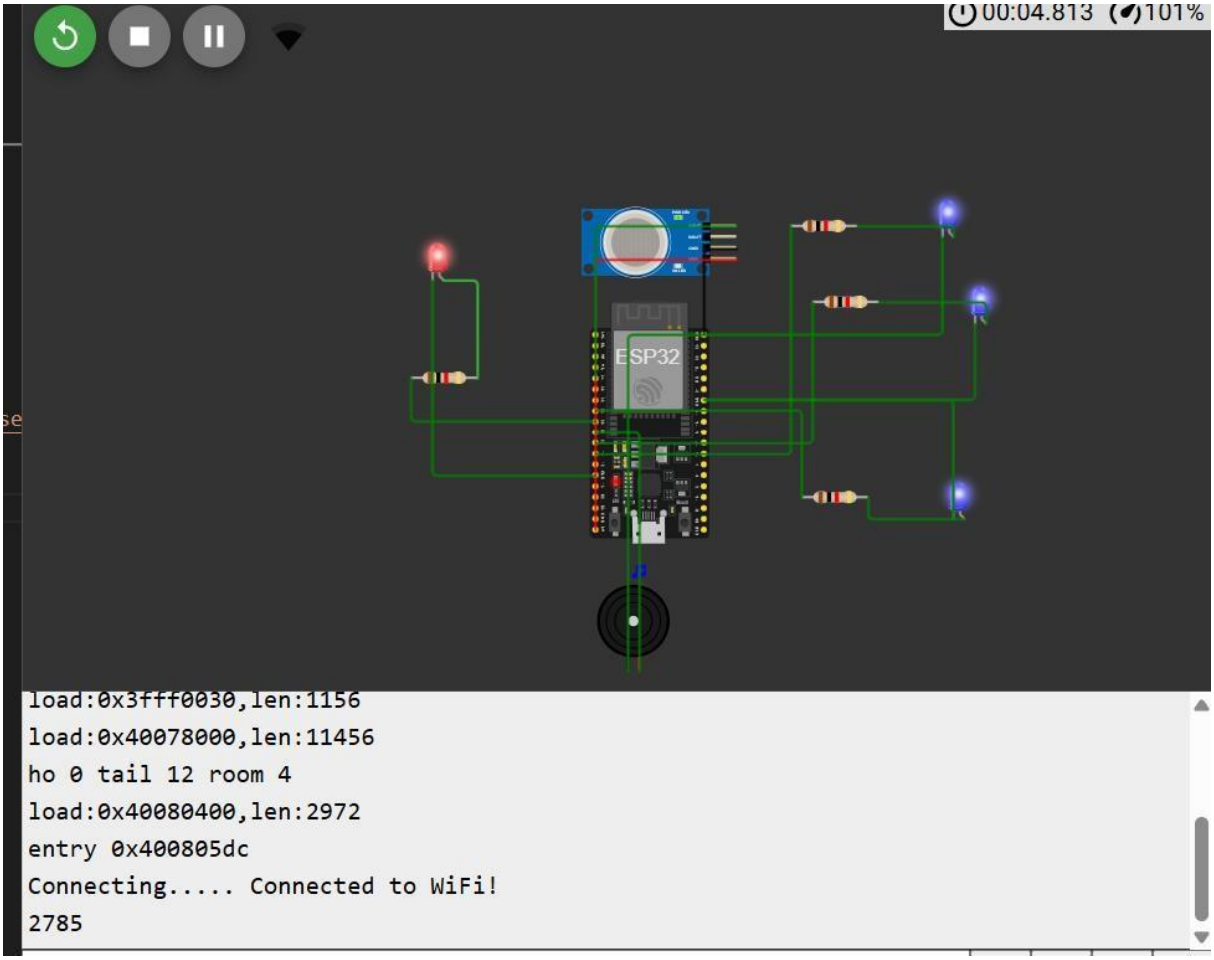
1. The wind sensor continuously monitors the surrounding wind intensity.
2. The sensed wind data is transmitted to the ESP32 microcontroller for processing.
3. The ESP32 compares the received wind value with a predefined threshold level.
4. When wind intensity exceeds the threshold:
   • The system activates safety control.
   • Smart door and smart window move to closed state.
   • Alert indication (LED/buzzer) is activated.
5. When wind intensity returns below the threshold:
   • The system stops the alert indication.
   • Smart door and window return to normal monitoring condition.
6. The process repeats continuously to ensure automatic protection and real-time response.

10. Results



https://iotass2-eaf70-default-rtdb.asia-southeast1.firebasedatabase.app/

- fire_status: false
- temperature: 25



https://iotass2-eaf70-default-rtdb.asia-southeast1.firebasedatabase.app/

- fire_status: true
- temperature: 25

11. Advantages

• Automatic operation without human effort

• Protects home from environmental damage

• Real-time monitoring capability

• Wireless communication enabled

• Low power consumption

• Scalable and expandable design

12. Future Enhancements

• Integration of rain, temperature, and humidity sensors

• Mobile application for remote monitoring

• Cloud data storage and analytics

• Smart notification system

• Motorized real door and window control

13. Conclusion

The Smart Door and Window System successfully demonstrates an IoT-based automation solution for smart homes. By integrating wind sensors with ESP32 and smart output devices, the system ensures safety and convenience. The simulation proves that the system is reliable, efficient, and suitable for future smart home applications.